Customer :

TYPE : 207XT Series Relay

RD Review 2022.05.10 Release

Revised : 2017-01-06
Issued : 2016-12-29

■ Features

High temperature resistance relay for OBC application .
Contact rating 32A 277vac at T105C ambient temperature
Comply with RoHS-Directive 2011/65/EU.

■ Type List

Terminal style	Contact form	Insulation system	Designation (provided with)
			Flux tight
PCB terminal	1A (SPNO)	F	207XT-1AH-F-C M02

■ Ordering Information

207X	T	-	1A	H	-	F	-	C
1	2		3	4		5		6

1. 207X -- Basic series designation
2. T -- High temperature type
3. 1A -- Single pole normally open
4. H -- Contact material Ag alloy
5. F -- Class F
6. C -- Flux tight

■ Contact Rating

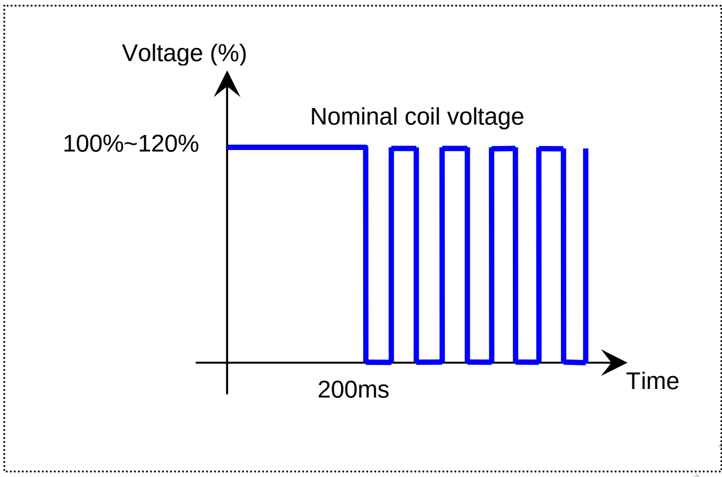
Resistive load	Making 1A, Carrying 32A, Breaking 1A / 240VAC, On 1s /Off 9s, at 105°C, 50K ops.

■ Coil Rating (DC)

Rated voltage (V)	Rated current ±10 % at 23°C (mA)	Coil resistance ±10 % at 23°C (Ω)	Pick up voltage (Max.) at 23°C (1)	Drop out voltage (Min.) at 23°C	Continuous voltage at 105°C (2)	Power consumption at rated / holding voltage
12	100	120	80 % of rated voltage	5 % of rated voltage	40~45 % of rated voltage	approx. 1.2W / 0.192W (2)

Notes : (1) To energize relay properly apply 100%~120% nominal coil voltage for 200ms.
(2) Coil holding voltage is 40~45% of nominal voltage after applying nominal voltage for 200ms.
(3) Recommend PWM method:
To energize relay properly apply 100%~120% nominal coil voltage for 200ms.
Start PWM control with duty ratio 40~45% and control frequency greater than 20 kHz.
Please see below figure 1.

Figure 1:



■ Specification

Contact material	Ag alloy	
Contact resistance [1]	100mΩ Max. (1 at 1A/6VDC by 4-wire resistance measurement) 10 mΩ Max. (By voltage drop 10A)	
Operate time [1]	15ms Max.	
Release time [1]	10ms Max.	
Insulation resistance [1]	100MΩ Min. (DC 500V)	
Dielectric strength [1]	Between open contact : AC 1000V, 50/60Hz 1 min.	
	Between contact and coil : AC 2500V, 50/60Hz 1 min.	
Vibration resistance	Operating extremes	10～50Hz , amplitude 1.0 mm
	Damage limits	10～50Hz , amplitude 1.0 mm
Shock resistance	Operating extremes	10G
	Damage limits	100G
Life expectancy	Mechanical	300,000 ops. (frequency 9,000 ops./hr)
Operating ambient temperature	-40～+105°C (no freezing)	
Weight	Approx. 15 g	

Note：(1) Initial value. Operate and release time excluding contact bounce.
(2) Unless otherwise specified, all tests are under room temperature and humidity.
(3) Consider the heat of PCB is necessary, please check the actual condition of PCB.
(4) Applying no diode to this relay. The life expectancy will be lower when a diode is used. To use a varistor (ZNR) could absorb the coil surge of relay that is recommended.
(5) Do not use the relay exceeding the coil rating, contact rating and life expectancy, or this may cause the risk of overheating.
(6) To assure optimum performance, avoid the relay from dropping, hitting, or other unnecessary shocks.
(7) Do not switch the contacts without any load as the contact resistance may become increased rapidly.
(8) Please contact Song Chuan for the detailed information.

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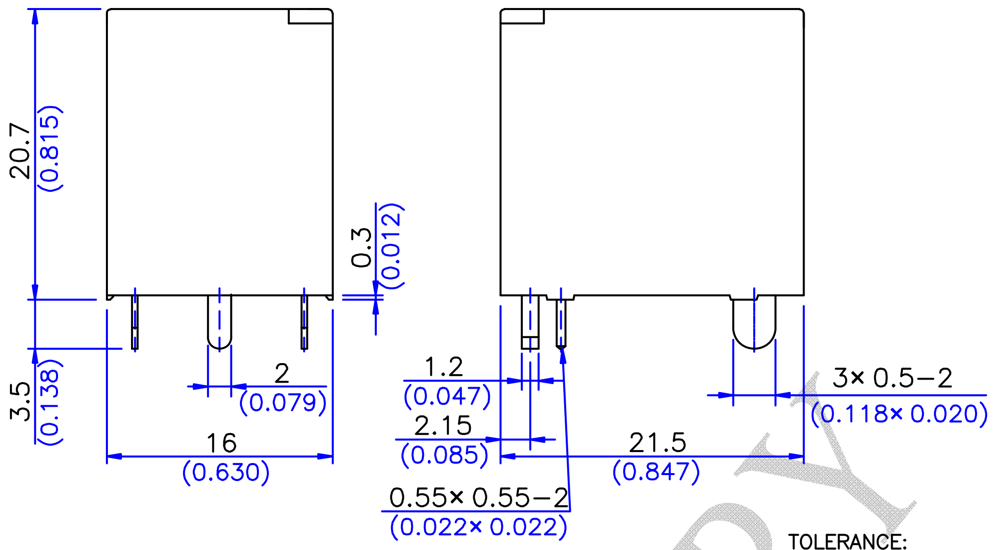
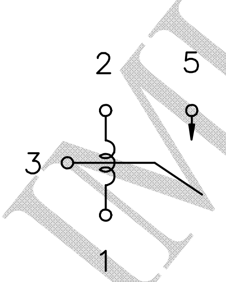
■ Outline Dimensions



TOLERANCE:
LESS THAN: 1(0.039)±0.1(0.004)
5(0.197)±0.3(0.012)
20(0.787)±0.5(0.020)
MORE THAN:20(0.787)±1(0.039)

■ Wiring Diagram
(Bottom view)



■ PC Board Layout
(Bottom view)